DE

3.2 What factors need to be considered when developing design solutions for manufacture?	
a. Awareness of the responsibilities and principles of total quality management (TQM), including: i. planning for accuracy and efficiency through testing and prototyping ii. being aware of issues in relation to different scales of production iii. designing for repair and maintenance iv. designing with consideration of product life.	

Considerations	*Maths & Science*
b. Awareness of issues related to product lifecycles that extend useful product life, such as: • products standing the test of time in terms of durability and style • maintenance and aftercare • re-working and recycling systems.	
c. Demonstrate an understanding of how environmental factors impact on: i. sourcing and processing raw materials into a workable form ii. the disposal of waste, surplus materials and components, by-products of production including pollution related to energy iii. cost implications related to materials and process.	
d. Demonstrate an understanding of sustainability issues relating to industrial manufacture, including: i. fair trade and the Ethical Trade Initiative (ETI) ii. economic issues and globalisation iii. material sustainability and optimisation, availability, recycling and conservation schemes, such as: ○ exploring the impact and use of eco-materials in the fashion/textiles chain ○ exploring how materials can be up-cycled.	